

Using random forest machine learning to predict plant geography and carbon fluxes

As an important component in the Earth system, vegetation provides links between the biosphere and atmosphere by regulating carbon, water, and energy fluxes. Therefore, vegetation develops in response to many different drivers, including climate, edaphic conditions, disturbances, and human activities. However, the distribution of vegetation is mostly controlled by climate types. This evolutionary convergence in community physiognomy (also called growth form composition) based on plant geography has been recognized as one of the oldest and greatest generalization of plant ecology.

The beginnings of predictive modelling in plant geography can be traced to Köppen's (1931) world climate zones. Köppen tried to match the distribution of biomes, and included relevant seasonal aspects of climate in his classification scheme. A later (but more artificial) classification scheme based on annual climate statistics by Holdridge (1947) (Figure 1) was used by Emanuel et al. (1985) to produce the first climate-derived map of global potential natural vegetation, and the first global projection of vegetation for a "greenhouse world" as simulated by a general circulation model (GCM, climate model).

Different research groups have made a lot of progress in understanding how terrestrial ecosystems change over time and space. They have combined the fields of plant geography, plant physiology and biogeochemistry, vegetation dynamics, and biophysics into the global dynamic vegetation model (DGVMs). Those models are now widely used to solve cutting-edge problems in the interactions between vegetation and climate.

Random forest models are often used as a classification tool to reconstruct land cover maps based on satellite or in situ observations, as they can account for nonlinear relationships between different variables. This project will use climate, soil, and biome observations and LPJ-GUESS (a DGVM developed at Lund University) outputs to make a predictive model of plant geography and carbon fluxes based on random forest algorithms. For the biome distributions, you can train observation data sets and LPJ-GUESS outputs and compare the differences between machine learning models and LPJ-GUESS outputs. For the carbon fluxes, you can use LPJ-GUESS outputs as the observation variables and test if the machine learning models can surrogate the LPJ-GUESS outputs or not.

Hands-on instructions:

We assume that only climate and soil type have an impact on the distribution of potential natural vegetation. We have provided such data sets for building a machine-learning model according to the following instructions:

1. Data

The gridded datasets (longitude: 0.5° , latitude: 0.5°) of climate, soil texture, biome type, topography, and carbon fluxes are available from Canvas.

To use these datasets, you should first read **Readme.txt**, in which information about variables are provided. In this folder, you will find "legend of biomes.txt", "Readme.txt", and a "requirements.txt" file for installing and running a python virtual environment locally. The datasets are located in the data-folder and comprises various variables, including statistical summaries such as mean, median, and standard deviation for climate variables across different seasons. These seasons are classified as spring, summer, autumn, and winter.

Additionally, the dataset contains soil texture data and LPJ GUESS output variables.

2. Data Description and Visualization

In this phase, provide an overview of the distribution of the outcome variables and select a few key features for visualization. Present these insights using graphical representations and a summary table of measures. You can refer to the RandomForestDemo notebook for inspiration.

Additionally, a starter notebook is provided where a visual representation of the biomes (Biome_obs) are showed on a global map.

3. Binary classification

In this part, you are going to build a binary classifier to classify two classes of biomes from "Biome_obs". You can choose one of the biomes and try to train a classifier that can distinguish this biome from the other data, e.g, 'Arctic/alpine tundra' or 'not Arctic/alpine tundra'. One way to narrow down the data if you have trouble creating the binary classifier is to only include samples from two different country codes or regions.

3.1. Data split

Split your dataset into two parts: a training set and a test set. The training set is used to train the model, and the test set is used to evaluate model performance. Think through how you split your data and what fits your purpose. Ways to split your data could be through climate, zone, biome, countries or continents. Are all the biomes represented in both your train and test data? Are the frequency of biome classes representative?

3.2. Model training

Next, train a Random Forest binary classification model. You can start by using training data from one smaller region (could be one country code). Evaluate your model and try to optimise the hyperparameters.

3.3. Model evaluation

Evaluate the model's performance using appropriate metrics for binary classification. Common metrics include precision, recall, and to plot the predictions in a confusion matrix. Test the model trained on one part of the world on the same biome in another part of the world, e.g., Arctic/alpine tundra in either North America or Siberia. Does the model show equal

skill in both areas? Could the features you have chosen affect the skill of the model in another part of the world? Why?

3.4. Feature importance and model interpretation

Analyze which features played the most crucial role in your model's decisions. Finally, discuss and interpret your results comprehensively, and explore any misclassifications to gain valuable insights into how well machine learning models adapt to different continents and environments. Are all features that you used for training necessary? Could you reduce the number of features used? Were there features that you did not use that could be important? Why would they be important?

Optional: Plot the misclassified regions on a world map.

4. Multiclass classification

In this part, you are going to build two multiclass random forest classification models to classify up to 18 biomes. You can use a combination of country code and region to build a subset that you as a group would be interested in, e.g. South America, Europe, South-east Asia et.c., again try to find another region where an interesting comparison can be made and if your model are performing well on a different geographic region. For example boreal Europe vs Canada, mainland USA (not including Alaska) vs. Australia or South America vs. Africa. **Important:** Use "Biome_obs" as your target variable.

4.1. Data split

Split your dataset into two parts: a training set, and a test set. The training set is used to train the model, the validation set to tune hyperparameters, and the test set to evaluate model performance.

4.2. Model Training

Next, train a Random Forest classification model using training data, and then put the model on the test data.

4.3 Hyper-parameter Tuning

Use cross-validation to tune hyperparameters like the number of trees, tree depth, and minimum leaf samples. This step can help improve model performance. If you encounter issues with runtime, consider narrowing your focus to fewer geographic regions. It is also possible to sample the grid list with the `.sample()` function on Pandas DataFrames, in that case, make sure to sample only the training dataset and not the full dataset.

4.4. Model Evaluation

Evaluate the model's performance using appropriate metrics for multiclass classification.

4.5. Features Importance

Analyze which features played the most crucial role in your model's decisions. What happens when you are limiting the number of features? Also, are there any auto-correlations in the feature data?

4.6. Compare LPJ-GUESS model and machine learning model

Compare your trained against the biomes estimated by LPJ-GUESS for “Biome_Cmax” and Biome_LAI. Discuss the difference in prediction between the LPJ-GUESS outputs and the machine learning model outputs. Where do the models agree and where do the model predictions diverge? Why?

5. Regression

In this part, you are going to build two random forest regression models to predict two continuous variables. One model to predict “NPP” and another to predict “VegC”. Split the data similar to section 4.1. and train your regression models on the training data.

5.1. Model Evaluation

Evaluate the model's performance using appropriate metrics for regression, such as mean squared error.

5.2. Feature Importance

Analyze which features played the most crucial role in your model's decisions.

5.3. Compare LPJ GUESS model and machine learning model

Discuss the difference in prediction between the LPJ GUESS model and the machine learning model. Just as in 4.6, it could be of interest to also analyse the geography of the performance of the model. An extension would be to use your trained biome-model as input to this model. Will that improve the results? And why? Another way of achieving this would be to create a model for each biome.

Hand-in

Write a short report (2-4 pages, including figures and references, standard A4 formatting with 2.5 cm margins and fontsize 12). In your report, include results from all three classifiers (binary, multiclass and regression). Describe the choices made during the training (for instance how you split your data), the most important features, and evaluation metrics. Discuss if the feature importance is reasonable. Also, discuss the differences in prediction between your model and LPJ-GUESS. Figures should be numbered and referred to in the text. The report should follow the structure (motivation/purpose, methods, results, discussion, conclusion, and reference). The report should be submitted to canvas.

AI use

Please write a small section on your use of AI (if any). This will not be included in your page limit or count towards your grade.

Evaluation criteria

Part	Criteria	Comments
Method	Method is well described for all elements of the task	
	Choices made during data preprocessing and model training are well described	
	Evaluation metrics are presented	
Analysis	Discussion of results shall show understanding of the topic	
	Suitability of evaluation metrics is discussed	
	Uncertainty and potential errors are discussed	
Presentation and sources	Correct citing of scientific literature	
	Keep to maximum length and correct formatting	
	General language and structure	